

Dr. Doc:

The nasal cavity is divided into two lateral halves by the nasal septum. Which cartilage contributes to the anterior part of this septum?

- A. Vomeronasal cartilage
- B. Alar cartilage
- C. Septal nasal cartilage
- D. Lateral nasal cartilage

The correct answer option is C. Septal nasal cartilage.

Explanation

C. Septal nasal cartilage: This is the correct choice because the septal nasal cartilage (or simply septal cartilage) forms the anterior, cartilaginous portion of the nasal septum. It fills the dynamic space in front of the bony components (the perpendicular plate of the ethmoid bone and the vomer) to separate the left and right airways.

Why the Other Options Are Incorrect

A. Vomeronasal cartilage: This is a tiny, rudimentary strip of cartilage located at the anterior-inferior margin of the septum, associated with the vestigial vomeronasal organ. It is not the primary cartilage forming the anterior septum.

B. Alar cartilage: These paired cartilages form the tip (apex) and the wings (ala) of the nostrils. While they flank the midline at the very tip, they do not form the main anterior structural wall of the internal nasal septum.

C. Lateral nasal cartilage: These are wing-like expansions extending laterally from the midline. They form the upper lateral walls and dorsum of the nose, rather than partitioning the cavity down the center.

In the bronchial tree, which structure is the first to lose cartilage support entirely?

- A. Tertiary bronchi
- B. Lobar bronchi
- C. Bronchioles
- D. Trachea

Correct answer:- C

- ▼ Tertiary bronchi still contain plates of hyaline cartilage to maintain their shape.
- ▼ Lobar bronchi have significant cartilage rings or plates.
- ▼ Bronchioles:- Cartilage is replaced by an increasing amount of smooth muscle at this level, allowing for bronchodilation and bronchoconstriction.
- ▼ The trachea is characterized by prominent C-shaped cartilage rings.

Which muscle elevates ribs 2 through 5 during inhalation?

- A. Sternocleidomastoid
- B. Serratus posterior inferior
- C. Serratus posterior superior
- D. Rectus abdominis

Correct Answer: C. Serratus posterior superior

Serratus posterior superior attaches to ribs 2–5 and elevates these ribs during inspiration ↑.

Think:

Superior = pulls ribs UP ↑

Ribs 2–5 = Serratus posterior superior

- A. Sternocleidomastoid — accessory muscle of inspiration; elevates the sternum, not specifically ribs 2–5.
- B. Serratus posterior inferior — primarily assists expiration by depressing the lower ribs.
- C. Serratus posterior superior — elevates ribs 2–5 during inspiration.
- D. Rectus abdominis — abdominal muscle; assists forced expiration.

Notices

Serratus posterior superior → ribs 2–5 → elevation → inspiration

Serratus posterior inferior → lower ribs → depression → expiration

Which functional zone of the respiratory system is responsible for filtering, humidifying, and warming incoming air?

- A. Respiratory zone
- B. Conducting zone
- C. Transition zone
- D. Alveolar zone

Answer: B. Conducting zone

Explanation:

- ▼ The conducting zone extends from the nose through the terminal bronchioles.
- ▼ Its major functions are conduction, filtration, warming, and humidification of inspired air.

▼ It does not normally participate in gas exchange and therefore constitutes part of the anatomic dead space.

▼ The respiratory zone begins with respiratory bronchioles and is specialized for gas exchange.

What type of epithelium lines the majority of the extrapulmonary region of the conducting portion?

- A. Simple squamous epithelium
- B. Stratified squamous epithelium
- C. Pseudostratified ciliated columnar epithelium
- D. Simple cuboidal epithelium

Answer: C. Pseudostratified ciliated columnar epithelium

Explanation:

▼ Most of the conducting respiratory tract is lined by pseudostratified ciliated columnar epithelium, commonly called respiratory epithelium.

▼ It contains ciliated cells, which propel mucus toward the pharynx, and goblet cells, which produce mucus.

🌀 Together they form the mucociliary escalator, an important defense mechanism against inhaled particles and microorganisms.

▼ The epithelium becomes progressively simpler as the airway diameter decreases.

What is the space between the true vocal folds called?

- A. Rima vestibuli
- B. Glottis (or Rima glottidis)
- C. Laryngeal ventricle
- D. Vestibule

Answer: B. Glottis (or Rima glottidis)

Explanation:

▼ The rima glottidis is the opening between the true vocal folds. The term glottis is often used to refer to the true vocal folds together with the opening between them. The vestibular folds lie superiorly and are not primarily responsible for normal voice production.

The epiglottis is unique among laryngeal cartilages because it is composed of:

- A. Hyaline cartilage
- B. Fibrocartilage
- C. Elastic cartilage
- D. Calcified cartilage

Answer: C. Elastic cartilage

Explanation:

▼ The epiglottis is composed primarily of elastic cartilage, allowing it to bend during swallowing and return to its resting position.

During swallowing, elevation of the larynx and backward movement of the epiglottis help protect the airway from aspiration.

Which cartilage forms the inferior base of the larynx and is ring-shaped?

- A. Thyroid
- B. Cricoid
- C. Arytenoid
- D. Corniculate

Answer: B. Cricoid

Explanation:

▼ The cricoid cartilage is the most inferior laryngeal cartilage and forms a complete ring around the airway.

▼ It is narrower anteriorly and broader posteriorly, giving it a signet-ring appearance. Inferiorly, it is continuous with the trachea through the cricotracheal ligament.

Which laryngeal cartilage is the largest and forms the "Adam's apple"?

- A. Cricoid cartilage
- B. Epiglottis
- C. Thyroid cartilage

D. Arytenoid cartilage

Answer: C. Thyroid cartilage

Explanation:

▼ The thyroid cartilage is the largest laryngeal cartilage. Its anterior projection forms the laryngeal prominence, commonly called the Adam's apple.

▼ It consists mainly of hyaline cartilage and provides structural protection and attachment for several laryngeal muscles and ligaments.

Which of the following is NOT a component of the thoracic wall?

A) Lungs

B) Skin

C) Fascia

D) Thoracic skeleton

Correct Answer: A

Explanation: The thoracic wall consists of skin, fascia, nerves, vessels, muscles, and the thoracic skeleton. The lungs are located within the thoracic cavity, not the wall.

Which structure is located within the thoracic cavity?

A) Distal part of the trachea

B) Cervical trachea

C) Thyroid gland

D) Pharynx

Correct Answer: A

Explanation: The thoracic cavity contains the heart, lungs, thymus, distal part of the trachea, and most of the esophagus.

What bones/bone groups comprise the skeleton of the thoracic wall?

- A) Sternum, humerus, and ribs
- B) Clavicle, scapula, and ribs
- C) Cervical vertebrae, ribs, and sternum
- D) Thoracic vertebrae, ribs, and sternum

Correct Answer: D

Explanation: The bones of the thorax include the thoracic vertebrae, ribs, and sternum.

What is located in the intercostal spaces that separate the ribs and their cartilages?

- A) Intercostal muscles, vessels, and nerves
- B) Lymphatic trunks and pulmonary arteries
- C) Vagus and phrenic nerves
- D) Deep cervical lymph nodes

Correct Answer: A

Explanation: The intercostal spaces separating the ribs contain intercostal muscles, vessels, and nerves.

How are ribs physically characterized in terms of shape?

- A) Curved, flat bones

- B) Curved, long cylindrical bones
- C) Short, irregular bones
- D) Thick, sesamoid bones

Correct Answer: A

Explanation: The ribs are curved, flat bones that form most of the thoracic cage

Which thoracic vertebra articulates with the first rib and the upper part of the second rib?

- A) First thoracic vertebra
- B) Second thoracic vertebra
- C) Seventh cervical vertebra
- D) Fifth thoracic vertebra

Correct Answer: A

Where do the cartilages of Floating ribs terminate?

- A) Central tendon of the diaphragm
- B) Posterior abdominal musculature
- C) Linea alba
- D) Infrasternal angle

Correct Answer: B

Explanation: The cartilages of the floating ribs end in the posterior abdominal musculature

Which angle of the inferior thoracic aperture is located at the xiphisternal joint?

- A) Subcostal angle
- B) Sternal angle of Louis
- C) Vertebrocostal angle
- D) Thoracic inlet angle

Correct Answer: A

Explanation: The xiphisternal joint is also the site of the infrasternal angle (subcostal angle) of the inferior thoracic aperture

The xiphoid process is a midline marker for the superior limit of which abdominal organ?

- A) Spleen
- B) Stomach
- C) Liver
- D) Gallbladder

Correct Answer: C

Explanation: The xiphoid process is a midline marker for the superior limit of the liver

The xiphoid process is a midline marker for which respiratory structure?

- A) Central tendon of the diaphragm
- B) Bifurcation of the trachea
- C) Apex of the right lung
- D) Left main bronchus

Correct Answer: A

Explanation: The xiphoid process is a midline marker for the superior limit of the central tendon of the diaphragm.

The xiphoid process is a midline marker for which cardiovascular border?

- A) Inferior border of the heart
- B) Superior border of the pericardium
- C) Arch of the aorta
- D) Bifurcation of the pulmonary trunk

Correct Answer: A

Explanation: The xiphoid process is a midline marker for the superior limit of the inferior border of the heart

The sternal angle of Louis marks the site of articulation of which rib with the sternum?

- A) Rib VII
- B) Rib I
- C) Rib V
- D) Rib II

Correct Answer: D

Explanation: The elevation at the manubriosternal joint (sternal angle) marks the site of articulation of rib II with the sternum

Why is Rib I not palpable on chest examination?

- A) It lies inferior to the clavicle and is embedded in base of neck tissues

- B) It is entirely cartilaginous and soft
- C) It is fused to the sternum and clavicle
- D) It lies deep to the scapula

Correct Answer: A

Explanation: Rib I is not palpable because it lies inferior to the clavicle and is embedded in tissues at the base of the neck.

Which rib serves as the standard anatomical reference for counting ribs?

- A) Rib III
- B) Rib I
- C) Rib II
- D) Rib VII

Correct Answer: C

The sternal angle lies at the level of which intervertebral space?

- A) Between vertebrae T4 and T5
- B) Between vertebrae T1 and T2
- C) Between vertebrae T8 and T9
- D) Between vertebrae C6 and C7

Correct Answer: A

Explanation: The sternal angle lies between vertebrae T4 and T5

The sternal angle marks which border of the pericardium?

- A) Superior border

- B) Inferior border
- C) Posterior border
- D) Lateral border

Correct Answer: A

Explanation: The sternal angle marks the superior border of the pericardium

The innermost layer of the intercostal muscles is split into which three muscles?

- A) Transversus thoracis (anterior), innermost intercostal (lateral), and subcostal (posterior)
- B) External intercostal, internal intercostal, and subcostal
- C) Pectoralis major, minor, and subclavius
- D) Sternomastoid, scalenus, and pectoralis minor

Correct Answer: A

Explanation: The innermost layer is split into the transversus thoracis (anteriorly), innermost intercostal (laterally), and subcostal (posteriorly).

Which of the following is NOT listed as a factor influencing the diaphragm's position?

- A) Gravitational force of the underlying viscera
- B) Elastic retraction force of the lungs
- C) Pressure in the abdominal cavity
- D) Systemic arterial blood pressure

Correct Answer: D

Explanation: The position of the diaphragm is influenced by the elastic retraction force of the lungs, pressure in the

abdominal cavity, and the gravitational force of the underlying viscera.

In which body position does the diaphragm operate at its lowest level?

- A) Sitting or standing
- B) Lying flat in a supine position
- C) Lying on the side (lateral decubitus)
- D) Inverted head-down position

Correct Answer: A

Explanation: When a person is sitting or standing, the diaphragm operates at its lowest level

At which vertebral level is the opening for the inferior vena cava located, and in what part of the diaphragm?

- A) Level of T8 vertebra; in the central tendon
- B) Level of T10 vertebra; in the muscular part
- C) Level of T12 vertebra; between the crura
- D) Level of T6 vertebra; in the muscular part

Correct Answer: A

Explanation: The opening for the inferior vena cava is at the level of the T8 vertebra, within the central tendon part.

Which structures are transmitted through the inferior vena cava opening of the diaphragm?

- A) Inferior vena cava and Right phrenic nerve
- B) Inferior vena cava and Vagus nerves
- C) Aorta, thoracic duct, and right phrenic nerve
- D) Esophagus and left gastric vessels

Correct Answer: A

Explanation: The inferior vena cava and the Right phrenic nerve are transmitted through the T8 opening in the central

tendon.

At which vertebral level is the esophageal hiatus located, and in what part of the diaphragm?

- A) At T12 vertebra; in the muscular part
- B) At T8 vertebra; in the central tendon
- C) At T10 vertebra; in the muscular part
- D) At T11 vertebra; between the crura

Correct Answer: C

Explanation: The esophageal hiatus is located at the T10 vertebra, in the muscular part of the diaphragm

Which of the following is transmitted through the esophageal hiatus?

- A) Esophagus, Right & Left vagus nerves, Left gastric vessels, and lymphatic vessels
- B) Esophagus, Right phrenic nerve, and thoracic duct
- C) Aorta, esophageal branches, and azygos vein
- D) Inferior vena cava and left gastric vessels

Correct Answer: A

Explanation: The esophageal hiatus transmits the esophagus, the Right and Left vagus nerves, the Left gastric vessels (esophageal branches), and lymphatic vessels.

At which vertebral level is the aortic hiatus located, and where?

- A) At level of T12 vertebra; in the muscular part between the crura

- B) At level of T10 vertebra; in the central tendon
- C) At level of T8 vertebra; in the muscular part
- D) At level of L1 vertebra; in the central tendon

Correct Answer: A

Explanation: The aortic hiatus is located at the level of the T12 vertebra, in the muscular part between the crura.

Which structures are transmitted through the aortic hiatus of the diaphragm?

- A) Aorta, Thoracic duct, and Azygos vein
- B) Aorta, Esophagus, and Vagus nerves
- C) Inferior vena cava, Aorta, and Right phrenic nerve
- D) Aorta, Sympathetic trunks, and Thoracic duct

Correct Answer: A

Explanation: The structures transmitted through the aortic hiatus are the aorta, the thoracic duct, and the azygos vein

Opposite to which thoracic vertebrae does the root of the lung lie?

- A) Vertebrae C6, C7, and T1
- B) Vertebrae T1, T2, and T3
- C) Vertebrae T8, T9, and T10
- D) Vertebrae T5, T6, and T7

Correct Answer: D

Explanation: The root of the lung lies opposite thoracic vertebrae T5, T6, and T7.

What is the root/hilum of the lung covered by?

- A) A sleeve of parietal pleura
- B) A layer of buccopharyngeal fascia
- C) Visceral pericardium
- D) Cervical pleura

Correct Answer: A

Explanation: The root/hilum of the lung is covered by a sleeve of parietal pleura

Which major vein runs posterior to the root of the Right lung?

- A) Azygos vein
- B) Hemiazygos vein
- C) Inferior thyroid vein
- D) Internal jugular vein

Correct Answer: A

Explanation: The root of the Right lung has the azygos vein looping over or running posterior to its structures.

The trachea lies in the midline except at its bifurcation where it slightly deviates to which side?

- A) To the left
- B) To the Right
- C) Anteriorly
- D) Posteriorly

Correct Answer: B

Explanation: The trachea lies in the midline except at its bifurcation where it slightly deviates to the right

Which structure lies directly anterior to the 2nd, 3rd, and 4th rings of the trachea?

- A) Left innominate vein
- B) Esophagus
- C) Recurrent laryngeal nerve
- D) Isthmus of thyroid gland

Correct Answer: D

Explanation: Anteriorly, the isthmus of the thyroid gland lies in front of the 2nd, 3rd, and 4th rings of the trachea

Which vascular structures lie anterior to the trachea below the thyroid isthmus?

- A) Inferior thyroid vein and thyroid ima
- B) Superior thyroid artery and brachiocephalic vein
- C) Left innominate vein and jugular arch
- D) Carotid sheath and internal jugular vein

Correct Answer: A

Explanation: Below the thyroid isthmus, the inferior thyroid vein and thyroid ima (if present) are located anterior to the trachea.

What are the lateral relations of the cervical portion of the trachea?

- A) Lobe of thyroid gland and carotid sheath with its contents
- B) Esophagus and recurrent laryngeal nerve
- C) Longus coli and vertebral column
- D) Left innominate vein and jugular arch

Correct Answer: A

Explanation: Laterally, the cervical trachea is related to the lobe of the thyroid gland and the carotid sheath and its content.

What is the arterial supply of the cervical portion of the trachea?

- A) Branches from inferior thyroid artery
- B) Branches from superior thyroid artery
- C) Branches from internal thoracic artery
- D) Direct branches from the subclavian artery

Correct Answer: A

Explanation: The arterial supply to the cervical portion of the trachea is by branches from the inferior thyroid artery

Which lymph nodes drain the cervical portion of the trachea?

- A) Submandibular lymph nodes
- B) Deep cervical lymph nodes
- C) Retropharyngeal lymph nodes
- D) Pretracheal and paratracheal lymph nodes

Correct Answer: A

Explanation: The lymphatic drainage of the cervical portion of the trachea goes to the pretracheal and paratracheal lymph nodes.

Where is tracheostomy surgery usually performed in relation to the thyroid isthmus?

- A) Superior to the thyroid isthmus
- B) Inferior to the thyroid isthmus
- C) Directly through the thyroid isthmus
- D) Lateral to the thyroid isthmus

Correct Answer: B

Explanation: Clinical note: tracheostomy surgery is usually performed inferior to the thyroid isthmus

Dr. Doc:

What is the anatomical position of the larynx relative to the laryngopharynx?

- A) Directly behind the laryngopharynx, opposite C1 to C3
- B) In front of the laryngopharynx, opposite C3 to C6
- C) Lateral to the laryngopharynx, opposite C7 to T2
- D) Superior to the laryngopharynx, opposite C1 to C2

Correct Answer: B

Explanation: larynx is located at the midline of the neck, running from the root of the tongue to the trachea. It is situated 'in front of laryngopharynx opposite C3 to C6' vertebrae.

How are the 9 cartilages of the laryngeal framework held together and interconnected?

- A) Directly fused by bone tissue
- B) Joined by ligaments and membranes
- C) Suspended purely by skeletal muscles without direct connective tissues

D) Connected solely by synovial cartilage sheets

Correct Answer: B

Explanation: the larynx contains '9 cartilages joined by ligaments and membranes' which allow them to function as a unified protective and vocal framework.

Which cartilage of the larynx is unpaired?

- A) Arytenoid cartilage
- B) Cricoid cartilage
- C) Corniculate cartilage
- D) Cuneiform cartilage

Correct Answer: B

Explanation: The three unpaired cartilages are the thyroid, cricoid, and epiglottis. The cricoid cartilage is therefore unpaired. The arytenoid, corniculate, and cuneiform are all paired cartilages.

Which cartilage of the larynx is paired?

- A) Thyroid cartilage
- B) Epiglottis
- C) Arytenoid cartilage
- D) Cricoid cartilage

Correct Answer: C

Explanation: The three paired cartilages are the arytenoid, corniculate, and cuneiform. Therefore, the arytenoid cartilage is a paired cartilage. Thyroid, cricoid, and epiglottis are unpaired.

Which cartilage of the larynx is described as ring-shaped, having a quadrilateral lamina posteriorly

and a narrow arch anteriorly?

- A) Thyroid cartilage
- B) Epiglottis
- C) Cricoid cartilage
- D) Arytenoid cartilage

Correct Answer: C

Explanation: cricoid cartilage—> 'Ring shaped, having quadrilateral lamina posteriorly and narrow arch anteriorly'

Which part of the cricoid cartilage projects upwards and articulates with the arytenoid cartilages?

- A) The narrow arch anteriorly
- B) The pointed lower end
- C) The quadrilateral lamina posteriorly
- D) The oblique line

Correct Answer: C

The transverse process of the C6 vertebra is a site where which major blood vessel can be compressed against it at the cricoid level?

- A) Internal jugular vein
- B) Subclavian artery
- C) Carotid artery
- D) Vertebral artery

Correct Answer: C

Explanation: The cricoid cartilage indicates the 'Site where the carotid artery can be compressed against the transverse process of the C6 vertebra'.

Which nerve enters the larynx at the level indicated by the cricoid cartilage?

- A) Internal laryngeal nerve
- B) Superior laryngeal nerve
- C) Recurrent laryngeal nerve
- D) Glossopharyngeal nerve

Correct Answer: C

Explanation: The cricoid cartilage serves as a landmark indicating the 'Point where the recurrent laryngeal nerve enters the larynx'.

The epiglottis is anatomically described as a leaf-shaped, broad flat plate composed of what material?

- A) Hyaline cartilage
- B) Fibrocartilage
- C) Calcified bone
- D) Dense regular white fibrous tissue

Correct Answer: B

The broad, free upper end of the epiglottis projects behind which two structures?

- A) Trachea and thyroid gland
- B) Tongue and hyoid bone
- C) Arytenoid and cricoid cartilages
- D) Cervical vertebrae C3 and C4

Correct Answer: B

Explanation: The 'upper end – broad, free, projects behind the tongue and hyoid bone, above and in front of the inlet of larynx'

To what structure and by what means is the pointed lower end of the epiglottis attached?

- A) To the hyoid bone by the hyoepiglottic ligament

- B) Above the angle of fusion of the thyroid cartilage by the thyroepiglottic ligament
- C) To the cricoid cartilage by the cricothyroid ligament
- D) To the tongue by the glossoepiglottic fold

Correct Answer: B

Explanation: The lower end – pointed, attached above the angle of fusion of the thyroid cartilage by the thyroepiglottic ligament'.

The anterior surface of the epiglottis is connected to the tongue by which structure?

- A) Hyoepiglottic ligament
- B) Glossoepiglottic fold
- C) Aryepiglottic fold
- D) Thyroepiglottic ligament

Correct Answer: B

The anterior surface of the epiglottis is connected to the hyoid bone by which structure?

- A) Hyoepiglottic ligament
- B) Thyrohyoid membrane
- C) Median thyrohyoid ligament
- D) Glossoepiglottic fold

Correct Answer: A

 The anterior surface is connected to tongue by glossoepiglottic fold and hyoid by hyoepiglottic ligament

With which structure does the curved posteromedial apex of the arytenoid cartilage articulate?

- A) Thyroid cartilage

- B) Cuneiform cartilage
- C) Corniculate cartilage
- D) Hyoepiglottic ligament

Correct Answer: C

Explanation: for the arytenoid cartilage, the 'Apex – curved posteromedially and articulate with corniculate cartilage'.

Through what type of joint and with what structure does the base of the arytenoid cartilage articulate?

- A) Symphysis with the thyroid cartilage
- B) Synovial joint with the cricoid lamina
- C) Fibrous syndesmosis with the epiglottis
- D) Suture with the corniculate cartilage

Correct Answer: B

Which cartilages are primarily responsible for controlling the opening between the vocal cords, known as the rima glottidis?

- A) Cuneiform cartilages
- B) Arytenoid cartilages
- C) Thyroid cartilages
- D) Corniculate cartilages

Correct Answer: B

Explanation: the arytenoid cartilages 'control the opening between the vocal cords, the rima glottidis'.

What specific movements of the arytenoid cartilages are permitted by the crico-arytenoid joint?

- A) Only sliding and rotational movements
- B) Sliding, tilting, and rotary movements

- C) Only translation and axial translation
- D) Abduction and adduction with no rotation

Correct Answer: B

What movements are permitted by the cricothyroid joint?

- A) Sliding, tilting, and rotational movements
- B) Rotary and gliding movements
- C) Only tilting movements
- D) No movements, as it is a fixed joint

Correct Answer: B

The thickening of the posterior border of the thyrohyoid membrane is known as which of the following?

- A) Median thyrohyoid ligament
- B) Lateral thyrohyoid ligament
- C) Vestibular ligament
- D) Cricothyroid ligament

Correct Answer: B

Which muscle covers the outer surface of the thyrohyoid membrane?

- A) Cricothyroid muscle
- B) Thyrohyoid muscle
- C) Thyroepiglottic muscle
- D) Aryepiglottic muscle

Correct Answer: B

Which laryngeal structures pierce the thyrohyoid membrane?

- A) Recurrent laryngeal nerve and inferior thyroid vessels
- B) Superior laryngeal vessels and internal laryngeal nerve
- C) External laryngeal nerve and superior thyroid artery
- D) Glossopharyngeal nerve and ascending pharyngeal artery

Correct Answer: B

Why is the triangular cricothyroid ligament considered a critical anatomical landmark for emergency airway access (the 'soft spot')?

- A) Because it is completely devoid of vascularity and does not bleed.
- B) Because the airway is closest to the skin and most accessible at this location.
- C) Because it contains no nerve endings and is painless to incise.
- D) Because it connects directly to the bronchi, bypassing the trachea.

Correct Answer: B

What is the primary functional difference between the extrinsic and intrinsic muscles of the larynx?

- A) Extrinsic muscles control phonation, whereas intrinsic muscles control laryngeal elevation.
- B) Extrinsic muscles move the larynx as a whole during swallowing, whereas intrinsic muscles move individual laryngeal parts.
- C) Extrinsic muscles are voluntary, whereas intrinsic muscles are completely involuntary.
- D) Extrinsic muscles are supplied by the recurrent laryngeal nerve, whereas intrinsic muscles are supplied by cervical spinal nerves.

Correct Answer: B

The thyrohyoid muscle has a unique combined extrinsic action. What is it?

- A) It elevates the hyoid and depresses the thyroid cartilage.

- B) It depresses the hyoid bone and elevates the thyroid cartilage.
- C) It tenses the vocal cords and widens the laryngeal inlet.
- D) It rotates the cricoid cartilage medially.

Correct Answer: B

Which of the following is the ONLY muscle of the larynx responsible for abducting the vocal folds and opening the rima glottidis?

- A) Lateral crico-arytenoid
- B) Transverse arytenoid
- C) Cricothyroid
- D) Posterior crico-arytenoid

Correct Answer: D

Explanation: The 'Posterior crico-arytenoid muscle' as the 'Abductor of vocal folds'. It rotates the arytenoids, which abducts the vocal processes and opens the rima glottidis.

Which intrinsic laryngeal muscle acts as the primary tensor of the vocal folds, and what is its mechanism of action?

- A) Vocalis muscle; tenses the folds by directly shortening the vocal ligament.
- B) Cricothyroid muscle; tenses the vocal folds by tilting the thyroid cartilage anteriorly.
- C) Lateral crico-arytenoid; tenses the folds by medially rotating the arytenoids.
- D) Thyro-arytenoid; tenses the folds by pulling the arytenoid posteriorly.

Correct Answer: B

Define the terms 'Rima vestibuli' and 'Rima glottidis'.

- A) Rima vestibuli is the space between the vocal folds; Rima glottidis is the space between the vestibular folds.

B) Rima vestibuli is the space between the vestibular folds; Rima glottidis is the space between the vocal folds.

C) Rima vestibuli is the space above the epiglottis; Rima glottidis is the space below the cricoid.

D) Rima vestibuli is the space inside the laryngeal sacculi; Rima glottidis is the laryngeal ventricle.

Correct Answer: B

Describe the motor innervation of the intrinsic laryngeal muscles.

A) All intrinsic muscles are supplied by the internal laryngeal nerve.

B) All intrinsic muscles are supplied by the recurrent laryngeal nerve except the cricothyroid, which is supplied by the external laryngeal nerve.

C) All intrinsic muscles are supplied by the external laryngeal nerve.

D) All intrinsic muscles are supplied by the superior laryngeal nerve.

Correct Answer: B

What is the correct sensory innervation pattern of the laryngeal cavity above and below the vocal folds?

A) Internal laryngeal nerve below the vocal folds, recurrent laryngeal nerve above the vocal folds.

B) Internal laryngeal nerve above the vocal folds, recurrent laryngeal nerve below the vocal folds.

C) External laryngeal nerve above the vocal folds, internal laryngeal nerve below the vocal folds.

D) Glossopharyngeal nerve above the vocal folds, vagus nerve below the vocal folds.

Correct Answer: B

The posterior aspect of the superior pole of the left kidney is related to which ribs?

A. Rib X only

B. Rib XII only

C. Ribs XI and XII

D. Ribs IX, X and XI

Answer: C. Ribs XI and XII

The right kidney lies lower than the left because of the large liver on the right side.

Left kidney → higher → related to ribs XI & XII

Right kidney → lower → related mainly to rib XII

The posterior margin of the inferior thoracic aperture is:

A. Superior to the anterior margin

B. At exactly the same level as the anterior margin

C. Inferior to the anterior margin

D. Limited by the xiphisternal joint

Answer: C. Inferior to the anterior margin